Assignment

Important Instruction : Practice all the questions

1. A stereo camera system has a baseline of 0.20 m and a focal length of 1000 pixels. Four corresponding feature points have horizontal coordinates in the left and right images as A(450, 420), B(610, 570), C(760, 735), and D(900, 850).
   Calculate the disparity and depth of each point. Arrange the points from nearest to farthest and analyse how a ±3-pixel error in disparity would influence the accuracy of the estimated depths.
2. While combining or aligning images captured from different viewpoints, techniques such as homography, rectification, DLT, and RANSAC are commonly used. Explain any three of these techniques with suitable examples and mention their role in image alignment or multi-view vision
3. In a stereo camera system, corresponding points in two images are related through epipolar geometry. Explain the role of the fundamental matrix in establishing this relationship. Also describe how it can be estimated from corresponding image points.
4. Explain how computer vision can support criminal detection using CCTV or surveillance images. Also compare image processing and computer vision by considering how each would handle the same surveillance image.
5. Consider the image:
   a. Determine the complete output matrix when stride is one and no padding.
6. Compare orthogonal, Euclidean and affine transformations based on the geometric properties preserved by each. A rectangular document in an image appears rotated, stretched and skewed. Determine which of these transformations would be most appropriate for correcting the document and justify your answer.

7. Two images of the same planar surface are captured from different viewpoints. Explain the transformation that can relate corresponding points between these images and discuss the conditions under which this relationship is valid. Also explain its use in image stitching, perspective correction, document scanning and planar-object tracking.

8. Perform Histogram equalization for given image

5	3	-2
0	1	4
2	-1	0

9. Is it difficult to determine the actual distance of an object using only a single perspective image? Compare monocular and stereo approaches for estimating depth, with suitable examples.